TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Jonathan Dordick, Ph.D. Seok-Joon Kwon, Ph.D.	Email	dordick@rpi.edu kwons2@rpi.edu
Institute Name	Rensselaer Polytechnic Institute			
Address	CBIS, 110 8th Street, Troy, NY 12180			

2. Technology Overview

Title of Proposed Technology	Development of Glyco-Engineered Universal T Cells using a 3D Tumor Organoid and Spheroid Pillar Array/Well Platform			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio・Medical		Bio・Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Cell and Tissue Therapeutics	Bio-processes / Bio-Equipment	Bio-Analytical Instruments
Concept of Proposed Technology	Current off-the-shelf (universal) CAR-T therapies encounter two significant challenges: Graft-versus-Host Disease (GvHD) and Host-versus-Graft (HvG) rejection. CRISPR-Cas9 technology effectively addresses GvHD by eliminating the T-cell receptor (TCR), yet HvG rejection mediated by the patient's natural killer (NK) cells and macrophages remains a critical obstacle.[1] We propose to advance next-generation universal T cell design using high-throughput combinatorial knockdown of specific glycosyltransferase genes, thereby enabling strategic modification of the T-cell surface glycan profile via glyco-engineering.[2] Specifically, in partnership with a Korean SME, we will use 3D Tumor Organoid and Spheroid Micropillar/Microwell Array Platforms[3-4] for high-throughput, three-dimensional functional screening of gene-edited T cells within solid tumor microenvironments.			
Rationale for Joint R&D Project	Overcoming the physical and immunological barriers of the solid tumor microenvironment remains the single greatest challenge in democratizing allogeneic cell therapies. We have strategically positioned ourselves to solve this by engineering a proprietary 3D micropillar array platform that allows for the high-content simultaneous assessment of T cell infiltration and cytotoxicity. To			

	advance this technology from the bench to the clinic, we specifically seek a Korean SME industry partner with the capability to secure patient-derived T cells and expertly manage patient-derived 3D tumor organoid microarrays. This collaboration is the final essential step to scale our platform, establishing a new global benchmark for high-throughput immuno-oncology screening and unlocking the discovery of innovative, next-generation universal T cell candidates.
Key Words	1. Glyco-Engineered Universal T Cells 2. High-throughput, 3D functional screening 3. Tumor Organoid and Spheroid Micropillar/Microwell Array 4. CRISPR-based combinatorial gene knockdown 5. Tumor Organoid-T Cell Co-Culture Systems

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (5)
R&D Objectives	Standard Universal T cells (TCR/HLA knockout) exhibit susceptibility to natural killer (NK) cell-mediated lysis due to the absence of "self" markers and the presence of stress ligands. Additionally, their pronounced glycocalyx may impede efficient infiltration into solid tumors. Targeted modification of glycosyltransferases—the enzymes responsible for synthesizing glycan chains—provides a potential strategy to address these limitations as follows. **(1) Evading NK Cell Recognition** to minimize immune clearance by eliminating specific ligands, particularly those related to hyper-sialylation. **(2) Enhancing Tumor Infiltration** by reducing physical obstacles such as mucin to improve access to the tumor microenvironment (TME). **(3) Increasing Therapeutic Potency** by altering the glycosylation patterns of immune checkpoints, including PD-1, to mitigate T-cell exhaustion and augment antitumor efficacy. In summary, our objective is to generate glyco-engineered universal T cells by employing CRISPR-based combinatorial knockdown and knockout strategies (see **Figure 1** below). By bypassing the lengthy manufacturing process, glyco-engineered allogeneic T cells can serve as an immediate, “off-the-shelf” intervention for patients facing aggressive, late-stage solid tumors (see **Figure 2** below). A series of tasks have been established as follows to advance this concept from TRL 3 to TRL 5. • Task 1. Combinatorial Library Construction (The "Glyco-Code") • Task 2. High-Throughput and High-Content Screening • Task 3. Optimization of Multiplex Glyco-Engineering in Primary T Cells • Task 4. Comprehensive *in Vitro* Validation of Universal Compatibility and Safety			

(max 10 pages)

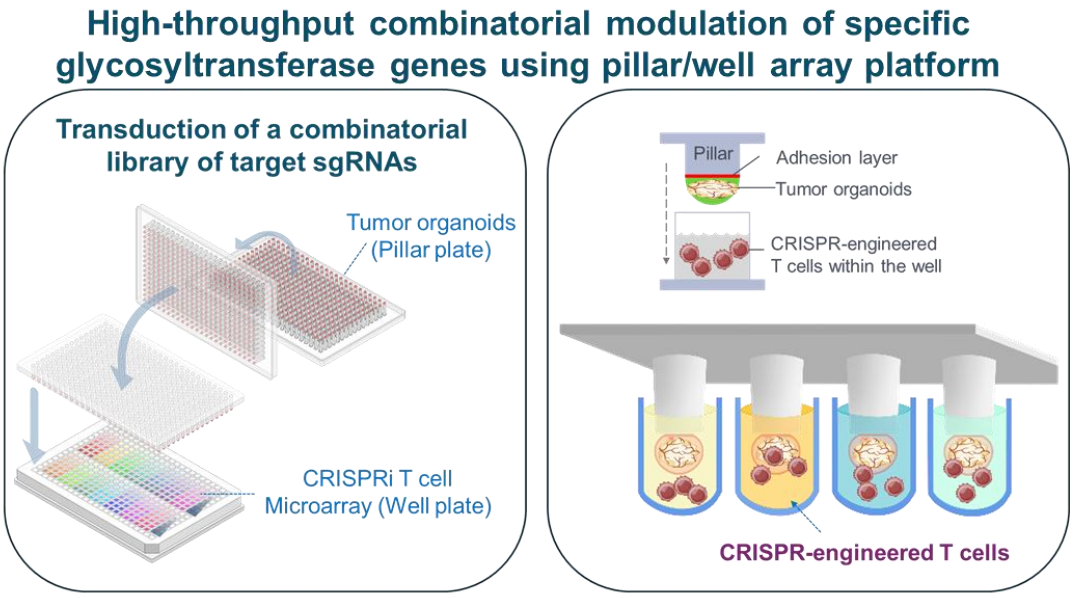

Figure 1. Schematic illustration of the high-throughput combinatorial screening process. A combinatorial sgRNA library is transduced into T cells to modulate specific glycosyltransferase genes. Using the 3D Pillar/Well array platform, CRISPR-engineered T cells in the wells interact with tumor organoids immobilized on the pillars. This setup facilitates the high-throughput identification of optimal gene combinations that enhance tumor infiltration and cytotoxicity.

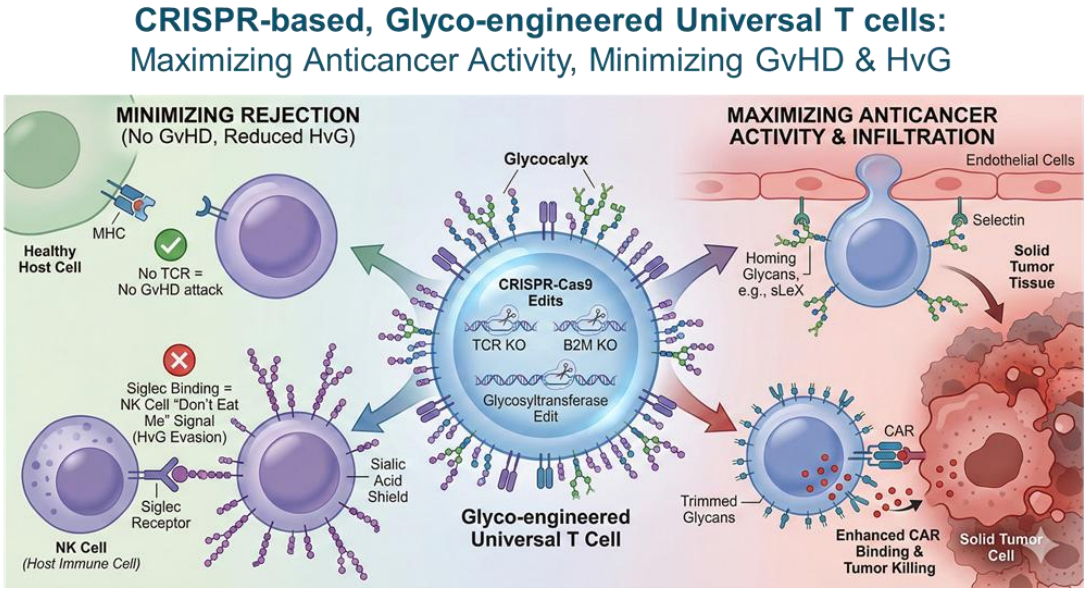

Figure 2. Schematic representation of the mechanism of action of CRISPR-based, glyco-engineered universal T cells. Mechanisms of maximizing efficacy and minimizing rejection: The engineered universal T cells undergo CRISPR-mediated knockout of TRAC and B2M to prevent Graft-versus-Host Disease (GvHD) and rejection by host T cells. Concurrently, glyco-engineering remodels the cell surface glycocalyx to achieve dual benefits: 1) Minimizing Rejection (**Left**): Interaction between the engineered sialic acid shield and inhibitory Siglec receptors on host NK cells prevents Host-versus-Graft (HvG) rejection ("Don't eat me" signal). 2) Maximizing Anticancer Activity (**Right**): Modulation of homing glycans (e.g., sLeX) promotes extravasation via interaction with endothelial Selectins, thereby maximizing infiltration into solid tumor tissues.

(max 10 pages)

R&D Target Specifications	1. Library Construction & Vector Production (**Tasks 1 and 2**)
	• Target Genes: Sialyltransferases (e.g. ST3GAL, ST6GAL), Fucosyltransferases (e.g. FUT8), Neuraminidases (NEU), etc. (Total ~20 candidate genes)
	• sgRNA Coverage: 3-4 distinct sgRNAs per target gene to ensure functional redundancy.
	• Lentiviral Titer: Functional titer ≥ 1.0×10^7 TU/mL (Transducing Units) for efficient primary cell transduction.
	• Cloning Efficiency: Sequence-verified correct insertion rate ≥90% for the multiplex library.
	2. High-Throughput 3D Screening (**Task 2**)
	• Platform: 384-Pillar/Well Array.
	• 3D Spheroid/Organoid Uniformity: Coefficient of Variation (CV) of spheroid/organoid diameter < 15% (to ensure reproducible screening data).
	• Hit Selection Criteria:
	  ○ Infiltration: > 2-fold increase in penetration depth compared to mock control.
	  ○ Killing Efficacy: Tumor volume reduction ≥ 40% under NK cell co-culture stress conditions.
	3. Multiplex Engineering of Primary T Cells (**Task 3**)
	• Cell Source: Patient-derived PBMCs (Activated T cells).
	• Multiplex Transduction Efficiency:
	  ○ dCas9-KRAB expression ≥ 50% of total population.
	  ○ Co-transduction (Vector A + Vector B) efficiency ≥ 40%.
	• Gene Silencing Efficiency (CRISPRi): Target mRNA knockdown ≥60% (measured by qPCR).
	• sLeX Induction: Surface sLeX expression level (MFI) adjustable to Low, Medium, High ranges (spanning 1-log dynamic range) via FUT7 titration.
	4. Universal Safety & Efficacy Validation (**Task 4**)
	• Universal Knockdown Efficiency:
	  ○ Surface TCR expression (CD3 negative) < 20% (to prevent GvHD).
	  ○ Surface HLA-I expression (B2M negative) < 30% (to prevent host T-cell rejection).
	• *In Vitro* Safety (GvHD): IFN-γ secretion in Mixed Lymphocyte Reaction (MLR) comparable to baseline background (No significant alloreactivity).
	• Immune Evasion (HvG): Survival rate against allogeneic NK cell cytotoxicity ≥ 50%.
	• Tumor Infiltration (3D Model): T cells detectable in the inner core (center) of the tumor spheroid.

R&D Plan	**Task 1. Combinatorial Library Construction (The "Glyco-Code") (Months 0-6, TRL 3).** Recombinant lentiviruses will be produced for combinatorial CRISPRi transduction using a micropillar and microwell array platform, consistent with our previously established protocols.[5-6] To achieve this, a collection of sgRNAs will be designed to target glycosyltransferase genes. These include genes associated with sialylation, such as ST3GAL1 and ST6GAL1, which can be modified to influence Siglec binding.[7] Additionally, genes involved in N-glycan branching, such as MGAT5, will be targeted to minimize the structural complexity of the glycocalyx.[8] Genes responsible for O-glycan elongation, such as C1GALT1, will also be targeted to reduce the length of mucin-like domains. Proper fucosylation ensures the T cell is sensitive enough to detect cancer cells but not so reactive that it

triggers autoimmune destruction. FUT7 (fucosyltransferase) is required for creating Sialyl Lewis X (sLeX) on T cells, which enables them to bind E-selectin on blood vessels.[7] This binding helps T cells migrate through vessel walls to tumors.[10] FUT8 can alter the physical spacing and organization of TCRs.[11]

Deliverable: A full proposal prepared jointly by RPI and a Korean SME.

Milestone: Bioinformatics design of sgRNAs targeting key immune-modulating genes, including sialyltransferases (ST3GAL1, ST6GAL1), N-glycan branching enzyme (MGAT5), O-glycan elongation enzyme (C1GALT1), and others.

Task 2. High-Throughput and High-Content Screening Using Immortalized T cells (Months 7-18, TRL 3).

To generate the screening library, recombinant lentiviral vectors, will be designed in Task 1 to deliver combinatorial payloads, are constructed with specific sgRNA libraries for CRISPRi-mediated silencing alongside expression cassettes for glycosyltransferases. These vectors are packaged into high-titer lentiviral particles using HEK293T cells and subsequently used to transduce immortalized T cells. This process yields a vast, diverse library of T cells, each characterized by a unique, genetically engineered surface glycan profile. The evaluation of these engineered cells relies on the innovative "Pillar-dips-into-Well" array platform, which serves as a high-content screening environment. On one side, human tumor spheroids or organoids are placed in Matrigel droplets on a Micropillar Array Plate. On the other side, a library of combinatorially glyco-edited immortalized T cells (e.g., Jurkat T cells) can be allocated to corresponding wells of a MicroWell Array Plate. By turning the pillar plate upside down and immersing it into the well plate, direct contact between the two components is achieved. This arrangement simulates the way T cells infiltrate and attack a solid tissue mass from its outer layers inward under natural conditions. To intensify testing and mimic a rejection scenario, either human NK cells or purified Siglec-Fc fusion proteins can be added to the MicroWells during the experiment. In these conditions, only T cells with appropriate glyco-modifications can withstand NK-induced stress and effectively destroy the tumor cells positioned on the pillar. Combinatorial sgRNA/CRISPRi and FUT expression in T cells can be implemented using the previously established micropillar/microwell array platform.[4,6] Through this process, we can rapidly identify the most effective glyco-engineered T cells for killing target tumor spheroids or organoids.

Deliverable: Screening data package and candidate selection report.

Milestone: Completion of high-throughput 3D screening using immortalized T cells and selection of lead candidates

Task 3. Optimization of Multiplex Glyco-Engineering in Primary T Cells (Months 19-30, TRL 3-4).

Building upon the optimal sgRNA candidates identified during the high-throughput screening phase, this task focuses on translating these findings into a clinically relevant model using patient-derived primary T cells. To achieve this, we will construct a single multiplex lentiviral vector designed to simultaneously express dCas9-KRAB and the selected array of sgRNAs, enabling the synchronized repression of multiple target glycosyltransferases responsible for immune rejection. Concurrently, to

fine-tune tumor homing capabilities, these T cells will be co-transduced with varying titers of a lentiviral vector expressing FUT7. This titration strategy allows for the generation of T cells with a spectrum of sLeX surface densities, facilitating the identification of the "sweet spot" that maximizes tumor infiltration without compromising cell viability or specificity. The functional efficacy of these multiplex-engineered primary T cells will be validated using the 3D Pillar/Well platform, where their ability to penetrate and eradicate patient-derived tumor organoids will be directly assessed. This comprehensive validation step aims to establish the optimal protocol for manufacturing high-potency, glyco-engineered universal T cells suitable for clinical application.

Deliverable: Primary T-cell optimization protocol and efficacy report; Provisional patent prepared; Joint publication prepared.

Milestone: Validation of optimal glyco-engineering in patient-derived primary T cells.

Task 4. Comprehensive *In Vitro* Validation of Universal Compatibility and Safety (Months 24-36, TRL 5).

In this final validation phase, we aim to assemble the complete "Universal" profile by integrating the immunological safety switches with the optimized glyco-engineering strategy established in Task III. We will construct a dedicated lentiviral vector encoding sgRNAs specifically designed to knock down TRAC and B2M, thereby inducing the repression of the T cell receptor and HLA class I molecules, respectively. This vector will be co-transduced into patient-derived primary T cells alongside the optimized glyco-modulation vectors (the "sweet spot" condition identified in Task III). The resulting fully engineered T cells will then undergo rigorous in vitro testing to evaluate their potential for Graft-versus-Host Disease (GvHD) and Host-versus-Graft (HvG) rejection. To assess GvHD risk, Mixed Lymphocyte Reactions (MLR) will be performed by co-culturing the engineered T cells with mismatched allogeneic PBMCs to confirm their inability to initiate an immune attack. Conversely, to evaluate HvG rejection, the engineered cells will be challenged with allogeneic T cells and NK cells. This set of experiments will verify whether the simultaneous downregulation of B2M and the optimization of surface glycans effectively shield the therapeutic cells from both adaptive (T cell-mediated) and innate (NK cell-mediated) immune clearance, ensuring their persistence and safety as an off-the-shelf therapy.

Deliverable: Pre-clinical safety and universal compatibility data package; Korean and U.S. Patent applications filed; Joint publication submitted.

Milestone: Final *in vitro* validation of the universal glyco-engineered T cell.

Overall Deliverable: A Validated "Universal Glyco-Engineered T Cell" Platform & Pre-Clinical Data Package.

The ultimate deliverable of this joint R&D project is the establishment of a "First-in-Class" Universal T Cell Therapy Candidate optimized for solid tumor treatment. This candidate will be defined by a proprietary, "Glyco-Code," a specific combination of genetic modifications that enables the cell to evade host immune rejection (NK/T cell clearance) while actively penetrating dense tumor microenvironments. Accompanying this therapeutic candidate is a validated high-throughput 3D screening platform capable of predicting the efficacy of solid tumor immunotherapies.

(max 10 pages)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Rensselaer Polytechnic Institute (RPI) will serve as the lead technology developer and principal molecular analyst throughout the entire project duration. In the initial phase (**Tasks 1 & 2**), RPI's primary responsibility is to design and construct the comprehensive combinatorial CRISPR lentiviral library targeting key glycosyltransferase genes and to conduct the initial high-throughput screening using Jurkat T cells on the microwell/micropillar platform. As the project progresses to primary cell optimization (**Task 3**), RPI will continue to lead the molecular engineering efforts by refining vector designs based on screening hits and performing genomic validation to ensure precise multiplex editing in the partner's patient-derived samples. In the final validation phase (**Task 4**), RPI will be responsible for constructing and optimizing the specific TRAC and B2M knockdown vectors required for the "Universal" profile and will lead the comprehensive data analysis of the immunological safety assays (GvHD/HvG), ensuring that the observed clinical safety correlates with the engineered genetic profile.
Expected R&R of Korean SME Partner	The Korean SME partner will function as the clinical translation hub and exclusive provider of the screening platform and patient-derived biological resources. The SME's core contribution involves supplying the proprietary 384-Pillar/Well array platform and the associated technical support necessary for RPI to conduct their initial screens (**Task 2**). Crucially, because the research institute is restricted to immortalized cells, the SME will be solely responsible for all validation steps involving patient-derived primary T cells (**Tasks 3 and 4**). This includes isolating primary T cells from patient PBMCs and transducing them with the optimal viral vectors provided by RPI to fine-tune sLeX expression levels and validate the "sweet spot" for tumor homing. Finally, the SME will execute the rigorous preclinical efficacy and safety studies—including GvHD and HvG assays and cytotoxicity tests against patient-derived tumor organoids—to confirm that the engineered cells are safe and effective for potential clinical application.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological Breakthroughs in Next-Generation Immuno-Oncology.** The primary technological achievement of this project will be the establishment of a novel "Glyco-Engineered Universal T Cell" platform that overcomes the critical limitations of current solid tumor therapies. By successfully identifying a patentable "Glyco-Code," a specific combinatorial gene signature that modulates surface sugars, we expect to create a therapeutic cell that simultaneously prevents Graft-versus-Host Disease (GvHD), evades Host-versus-Graft (HvG) rejection by NK cells, and exhibits superior infiltration into dense solid tumor tissues. This represents a paradigm shift in cell engineering, moving beyond the simple expression of Chimeric Antigen Receptors (CARs) to the fundamental remodeling of the cell's physical and immunological interface (the glycocalyx). This "first-in-class" strategy will provide a scientifically rigorous solution to the "solid tumor paradox," where therapeutic cells often fail to penetrate the tumor microenvironment or are rapidly cleared by the host immune system. **Commercialization and Platform Validation.** From a commercial perspective, this collaboration will serve as a definitive industrial validation of the

Korean SME partner's proprietary 384-Pillar/Well array technology. By demonstrating the platform's unique capability to perform high-throughput, 3D functional screening for complex phenotypes like tumor infiltration and immune evasion, we will position it as a superior alternative to conventional 2D assays currently used in drug discovery. This validation is expected to accelerate the platform's entry into the global bio-instrumentation market, establishing it as a new industry standard for pre-clinical immuno-oncology screening. Furthermore, the intellectual property generated regarding the optimal gene editing combinations will form the basis for a high-value asset, suitable for licensing to major pharmaceutical companies or for launching a spin-off or joint venture dedicated to developing off-the-shelf cell therapies.

Broader Industrial and Economic Impact.

On a broader scale, this project aims to expand access to advanced cancer treatments by advancing the development of "off-the-shelf" (allogeneic) therapies. Current autologous CAR-T treatments are prohibitively expensive and suffer from long manufacturing lead times, limiting their accessibility to a small fraction of patients. By perfecting a universally compatible T cell that can be mass-produced and stored, this research has the potential to drastically reduce manufacturing costs and treatment delays. This transition from "custom-made" to "ready-made" therapeutics will not only alleviate the financial burden on healthcare systems but also ensure that life-saving advanced therapies are available immediately to patients with rapidly progressing solid tumors, thereby creating significant social value and improving global public health outcomes.

Team

Jonathan S. Dordick, Ph.D., is Institute Professor of Chemical and Biological Engineering at RPI, with a joint appointment in the Department of Biological Sciences. He also serves as Vice President for Strategic Alliances and Translation. He received his B.A. degree in Biochemistry and Chemistry from Brandeis University and his Ph.D. in Biochemical Engineering from the Massachusetts Institute of Technology. At RPI, he has served as the Vice President for Research, founding Co-Director of the Rensselaer-Mount Sinai Center for Engineering and Precision Medicine, Director of the Center for Biotechnology & Interdisciplinary Studies, and Department Chair. Dr. Dordick has made foundational contributions to protein science and enzyme technology, biomanufacturing of therapeutic proteins and nucleic acids, and drug discovery. Dr. Dordick has received numerous awards and honors, including election to the National Academy of Engineering and National Academy of Inventors, and receipt of numerous society awards. He has founded several companies, and published over 460 papers and is an inventor/co-inventor on over 50 patents.

Seok-Joon Kwon, Ph.D., is a Senior Research Scientist at RPI with two decades of expertise bridging molecular biology, protein engineering, glycobiology, and nanobiotechnology. His research focuses on developing high-throughput screening assays, CRISPR-based systems, and combinatorial methods for drug discovery. He has a portfolio of over 60 publications, including papers in prestigious peer-reviewed journals, book chapters, and patents.

(max 10 pages)

6. References

(1) Depil, S., Duchateau, P., Grupp, S. A., Mufti, G. & Poirot, L. 'Off-the-shelf' allogeneic CAR T cells: development and challenges. *Nat. Rev. Drug Discov.* **19**, 185–199 (2020).

(2) Wisnovsky, S., Möckl, L., Malaker, S. A., Pedram, K., Hess, G. T., Riley, N. M. et al. Genome-wide CRISPR screens reveal a specific ligand for the glycan-binding immune checkpoint receptor Siglec-7. *Proc. Natl Acad. Sci. USA* **118**, e2015024118 (2021).

(3) Kwon, S., Lee, D. W., Shah, D. A., Ku, B., Jeon, S. Y., Solanki, K. et al. High-throughput and combinatorial gene expression on a chip for metabolism-induced toxicology screening. *Nat. Commun.* **5**, 3739 (2014).

(4) Gopal, S., Kwon, S., Ku, B., Lee D. W., Kim J. & Dordick, J. S. 3D tumor spheroid microarray for high-throughput, high-content natural killer cell-mediated cytotoxicity. *Commun. Biol.* **4**, 893 (2021).

(5) Lin, J., Miao, J., Schaefer, K. G., Russell, C. M., Pyron, R. J., Zhang, F., Phan, Q. T., Solis, N. V., Liu, H., Tashiro, M., Dordick, J. S., Linhardt, R. J., Yeaman, M. R., King, G. M., Barrera, F. N., Peters, B. M. & Filler, S. G. Sulfated glycosaminoglycans are host epithelial cell targets of the Candida albicans toxin candidalysin. *Nat. Microbiol.* **9**, 2553–2569 (2024).

(6) Kwon, S. J., Lee, D. W., Shah, D. A., Ku, B., Jeon, S. Y., Solanki, K., Ryan, J. D., Clark, D. S., Dordick, J. S. & Lee, M.-Y. High-throughput and combinatorial gene expression on a chip for metabolism-induced toxicology screening. *Nat. Commun.* **5**, 3739 (2014).

(7) Stanczak, M. A., Siddiqui, S. S., Trefny, M. P., Thommen D. S., Kayluz F. B., Stephan G. et al. Self-associated molecular patterns mediate cancer immune evasion by engaging Siglecs on T cells. *J. Clin. Invest.* **128**, 4912–4923 (2018).

(8) Demetriou, M., Granovsky, M., Quaggin, S. & Dennis, J. W. Negative regulation of T-cell activation and autoimmunity by Mgat5 N-glycosylation. *Nature* **409**, 733–739 (2001).

(9) Knibbs, R. N., Craig, R. A., Natsuka S., Chang A., Cameron M., Lowe, J. B. & Stoolman, L. M. The fucosyltransferase FucT-VII regulates E-selectin ligand synthesis in human T cells. *J. Cell Biol.* **133**, 911–920 (1996).

(10) Aires, D. J., Yoshida, M., Richardson S. K., Bai M., Liu L., Moreno R. et al. T-cell trafficking plays an essential role in tumor immunity. *Lab. Invest.* **99**, 85–92 (2019).

(11) Okada, M., Chikuma, S., Kondo, T., Hibino, S., Machiyama H., Yokosuka T. et al. Blockage of core fucosylation reduces cell-surface expression of PD-1 and promotes anti-tumor immune responses of T cells. *Cell Rep.* **20**, 1017–1028 (2017).

(max 10 pages)

Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization